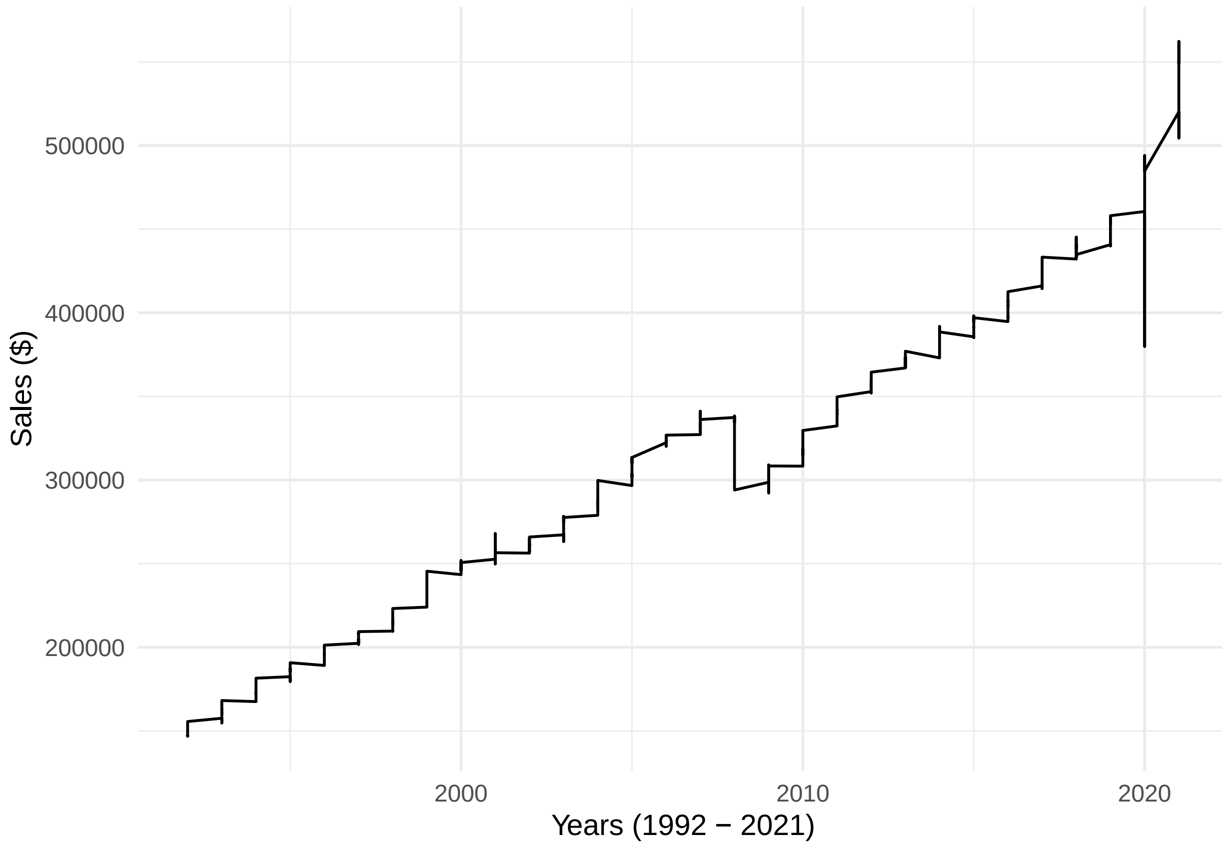
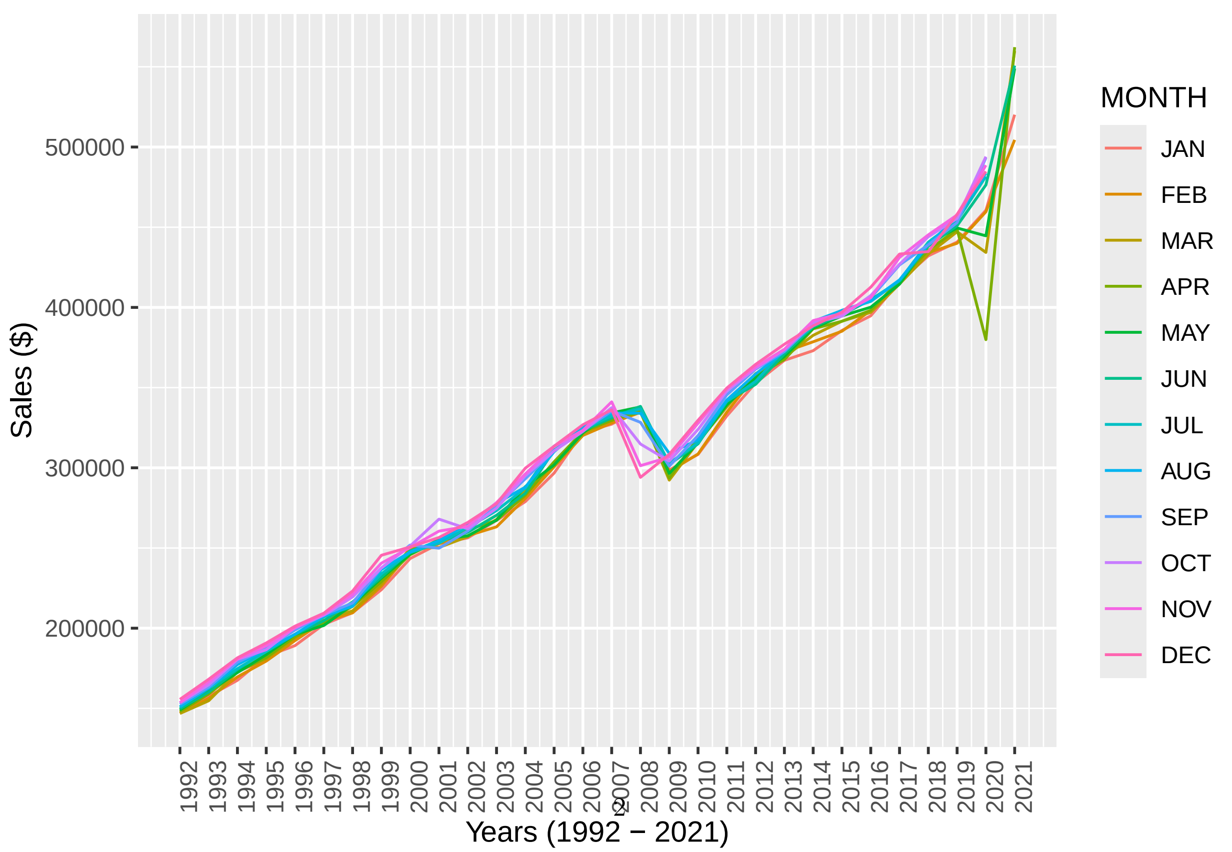
seth_dice_8.2

Seth Dice

2025-07-27

Plot the data with proper labeling and make some observations on the graph.





Looking at the time series graphs plotted we can see dips for the years 2007-2008 and 2020. The 2007-2008 dip can be explained by the market crash in '08. To explain the drop in 2020 we have to view the data split by months and we can see the 2020 dip started in April which would mark the closure of a lot of businesses in the US due to COVID.

Split this data into a training and test set. Use the last year of data (July 2020 – June 2021) of data as your test set and the rest as your training set.

Use the training set to build a predictive model for the monthly retail sales.

```
## Series: training_ts
## ARIMA(2,1,1) with drift
##
## Coefficients:
##          ar1      ar2      ma1     drift
##       0.0865  -0.4637  -0.2291  875.6044
## s.e.  0.1553   0.0760   0.1514  164.5520
##
## sigma^2 = 29606514:  log likelihood = -3415.32
## AIC=6840.64   AICc=6840.82   BIC=6859.8
```

Use the model to predict the monthly retail sales on the last year of data.

```
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## Jul 2020       448096.1 441122.9 455069.3 437431.6 458760.6
## Aug 2020       432155.6 422970.6 441340.6 418108.4 446202.8
## Sep 2020       445080.9 435518.8 454642.9 430456.9 459704.8
## Oct 2020       454795.5 444822.6 464768.5 439543.2 470047.9
## Nov 2020       450848.2 439952.8 461743.6 434185.1 467511.2
## Dec 2020       447208.3 435442.6 458974.0 429214.2 465202.3
## Jan 2021       449929.8 437588.6 462271.1 431055.5 468804.1
## Feb 2021       453058.8 440192.4 465925.1 433381.4 472736.1
## Mar 2021       453273.3 439810.5 466736.0 432683.8 473862.8
## Apr 2021       453046.9 438993.6 467100.1 431554.3 474539.5
## May 2021       454133.7 439553.0 468714.5 431834.4 476433.1
## Jun 2021       455538.6 440460.1 470617.1 432478.0 478599.1
```

Report the RMSE of the model predictions on the test set.

```
## [1] "The RMSE of model predictions on test set: 70370"
```